Paper review - Eyeriss

1. Dataflow

- 採用 Row Stationary(RS)的 dataflow，固定一個 PE row 的 filter weight，在每個 cycle 更新 ifmap value，或是兩者相反，可以用每個 PE 裡的 spad 存 filter weight、ifmap value 或是 psum。透過這種方式可以減少 access DRAM 或 GLB 的次數，以節約能源消耗。
  i. Convolutional Reuse：Each filter weight is reused $E \times F$ times in the same ifmap plane, and each ifmap pixel is usually reused $R \times S$ times in the same filter plane.
  ii. Filter Reuse：Each filter weight is reused across the batch of N ifmaps.
  iii. Ifmap Reuse: Each ifmap pixel is reused across M filters.
  iv. Psum accumulation：save psum in spad of each PE.
- 1-D convolution primitive in a PE：
  i. Each primitive operates on one row of filter weights and one row of ifmap values, and generates one row of psums.
  ii. Each PE can use the local spads for both convolutional data reuse and psum accumulation.
  iii. Required spad capacity depends only on the filter row size (S).
- 2-D Convolution PE Set：
  i. A 2-D convolution is composed of many 1-D convolution primitives.
  ii. Shares the same row of filter or ifmap across primitives.
  iii. Accumulates the psums from multiple primitives together.
  iv. Each row of filter is reused horizontally, each row of ifmap is reused diagonally, and rows of psum are accumulated vertically.
  v. The height and the width of the PE set are equal to the number of filter rows (R) and ofmap rows (E), respectively.
- PE Set Mapping：
  i. A PE set can be mapped to any group of PEs in the array that has the same dimensions.
  ii. If PE sets has more than 168 PEs, PE set can be strip-mined into $R \times e$,

where the PE set only processes *e* rows of ofmap at a time.

If PE set has less than 168 PEs, but has width larger than 14 or height larger than 12, a PE set that is too wide is divided into separated segments that are mapped independently to the array, but cannot support the mapping of a PE set that is taller than 12.

- Multiple 2-D Convolutions in a PE Set：
  i. By interleaving the computation of primitives that run on the same ifmap with different filters, the spads can buffer the same ifmap value and reuse it to compute with a weight from each filter sequentially.
  ii. By interleaving the computation of primitives that run on different channels, the PE can accumulate through all channels sequentially on the same psum.
- Multiple PE Sets in the PE Array：
  i. The PE array can fit more than one PE set if the set is small enough.
  ii. The PE array fits $r \times t$ PE sets in parallel that run r different channels of t different filters simultaneously.
  iii. Every t sets share the same ifmap with t filters.
  iv. Every r sets that run on r channels accumulate their psums.

Parameter	Description
m	number of ofmap channels stored in the global buffer
n	number of ifmaps used in a processing pass
e	width of the PE set (strip-mined if necessary)
p	number of filters processed by a PE set
q	number of channels processed by a PE set
r	number of PE sets that process different channels in the PE array
t	number of PE sets that process different filters in the PE array

2. Control flow

- Top-Level Control
  i. Read kernel size, ifmap, ofmap of CNN model, configure PE mapping.
  ii. traffic between the off-chip DRAM and the GLB through the asynchronous interface.
  iii. traffic between the GLB and the PE array through the NoC.
  iv. operation of the RLC CODEC and ReLU module.

v. Configure row and column IDs of NoC.

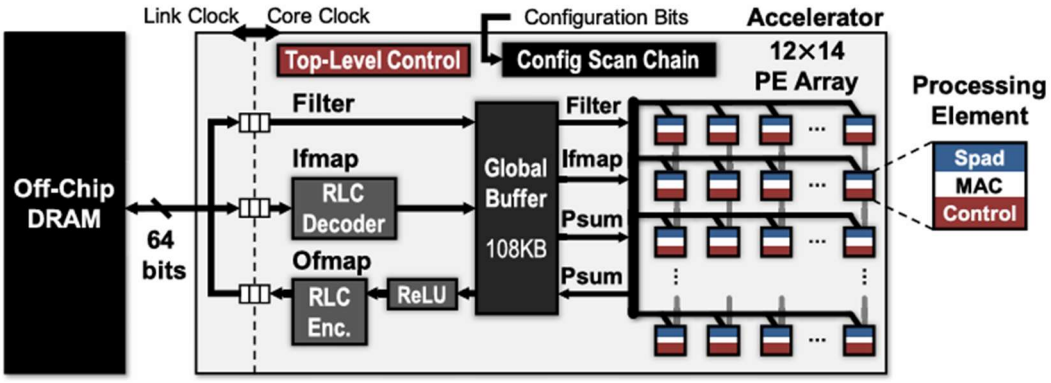
- PE-Level Control
  i. Control logic in each PE, which runs independently of each other.
  ii. Management of ifmap and filter weights.
  iii. Psum accumulation and propagation.
  iv. Data gating in PEs when encountering zero inputs.

3. Architecture

There are two clock domains that run independently and communicate through an asynchronous FIFO interface.

- Communication with the Off-Chip DRAM：
  i. 64-b bidirectional data bus.
- On-Chip processing：
  i. 168 PEs in the shape of 12 × 14 rectangle.
    A. Each PE contains a spad(local storage), MAC, and control logic.
    B. Pipelined into three stages：spad access, two-stage multiplier, and adder.
    C. Spads are separated for three data types：224-b × 16-b SRAM for filter spad, 12b × 16b register for ifmap spad, and 24b × 16b register for psum spad.
  ii. 108-kB GLB
    100 kB is allocated for ifmaps and psums, 8kB is allocated for filter weights.
  iii. RLC CODEC
    Since ReLU function introduces many zeros, the RLC is used to decompose consecutive zeros into 5-b Run, the next non-zero value is inserted as a 16-b Level. Every three pairs of run and level plus 1-b Term in the end are packed into a 64-b word.
  iv. ReLU module
  v. NoC
    A. Global Input Network(GIN) and Global Output Network(GON)：There are X-bus and Y-bus to compare with col ID and row ID.
    B. Local Network(LN)：A 64b data bus is implemented between

every pair of PEs that are on two consecutive rows of the same column.



4. Potential improvements

- Increasing Parallelism for Deeper CNN Layers

  In deep networks like VGG-16, ifmap become smaller in deeper layers, thus PE utilization decreases in these layer.

  Improvement：allow different layers to use different dataflows(Row Stationary or Weight Stationary).

- Mixed Precision Computation

  Eyeriss only operates using 16-b precision, but many modern models use quantized weights like 8-b, or even 4-b.

  Improvement：implement 8-b and 4-b MAC units, also apply dynamic precision configuration, allow layers to adjust precision based on the sensitivity of activations to accuracy loss.

- Supporting model parallelism

  Since modern models become larger and larger, it may require not only a single chip to accelerate.

  Improvement：implement chip-to-chip interconnects for multi-chip communication.